

# CodeTrain Milestone Assessment Rubrics

Grading criteria for Demo Days 1-4

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## 1 Using These Rubrics

Each Demo Day has a **25-point rubric**. Scores:

- **23-25:** Excellent (A)
- **20-22:** Very Good (B)
- **18-19:** Good (C)
- **15-17:** Pass (D)
- **<15:** Needs Improvement

Students must score **18+ to pass the milestone** and proceed to the next phase.

## 2 Mini Demo Day 1: Python Logic Script (Week 4)

**Goal:** Students demonstrate Python mastery through a working CLI tool.

### Submission Format:

- Working Python script ( `.py` file)
- README.md with usage instructions
- GitHub repository link
- 10-minute live demo + 5-minute Q&A

### Example Projects:

- MTN subscription manager (add/remove users, check balance)
- Voting system for school elections
- Simple expense tracker for Accra small businesses
- Student grade calculator

### 2.1 Scoring Rubric (25 points)

| Criterion    | Points | Excellent (5) | Good (3-4) | Needs Work (1-2) |
|--------------|--------|---------------|------------|------------------|
| Code Quality | 5      |               |            |                  |

| Criterion             | Points | Excellent (5)                        | Good (3-4)                 | Needs Work (1-2)           |
|-----------------------|--------|--------------------------------------|----------------------------|----------------------------|
|                       |        | Clean, Pythonic, good naming         | Mostly clean, minor issues | Hard to read, non-Pythonic |
| <b>Functionality</b>  | 10     | All features work perfectly, no bugs | Most features work         | Many bugs, incomplete      |
| <b>Error Handling</b> | 5      | Robust try/except for edge cases     | Some error handling        | No error handling          |
| <b>Documentation</b>  | 3      | Clear README, docstrings, comments   | Partial docs               | Minimal/no docs            |
| <b>Presentation</b>   | 2      | Clear demo, answers Q&A well         | Adequate demo              | Unclear or unprepared      |

**Minimum Score to Pass:** 18/25

## 3 Mini Demo Day 2: Exploratory Data Analysis Report (Week 8)

**Goal:** Students demonstrate Pandas mastery and data storytelling.

### Submission Format:

- Jupyter notebook ( `.ipynb` )
- PDF report (1-2 pages with key findings + 3 visualizations)
- 5-slide summary presentation
- GitHub repository link
- 10-minute live demo + 5-minute Q&A

### Required Analysis:

- Load and explore dataset (10+ data questions answered)
- 5+ publication-ready visualizations
- Identify and fix 3+ data quality issues
- Write 1-page executive summary with key insights

**Example Datasets:**

- Ghana student performance data (district, cohort, scores)
- MTN mobile money transactions (volume, growth, fraud patterns)
- Accra housing market (price trends, district comparisons)
- Ghana health clinic data (patient volume, disease prevalence)

**3.1 Scoring Rubric (25 points)**

| Criterion                          | Points | Excellent (5)                              | Good (3-4)              | Needs Work (1-2)         |
|------------------------------------|--------|--------------------------------------------|-------------------------|--------------------------|
| <b>Data Exploration</b>            | 5      | Thorough; answers 10+ meaningful questions | Adequate; 5-7 questions | Shallow; <5 questions    |
| <b>Visualizations</b>              | 8      | 5+ clear, labeled, insightful plots        | 3-4 adequate plots      | <3 plots or poor quality |
| <b>Data Cleaning</b>               | 5      | Handles missing, duplicates, outliers well | Handles some issues     | Minimal cleaning         |
| <b>Insights &amp; Storytelling</b> | 4      | Clear narrative, actionable findings       | Basic findings          | Vague or unclear         |
| <b>Presentation</b>                | 3      | Professional slides, clear demo            | Adequate presentation   | Disorganized or unclear  |

**Minimum Score to Pass:** 18/25

**4 Mini Demo Day 3: Predictive Model Build (Week 12)**

**Goal:** Students demonstrate ML competency: train, evaluate, and interpret models.

**Submission Format:**

- Saved trained model ( `.pkl` or `.joblib` )
- Jupyter notebook with full pipeline
- Evaluation report (2–3 pages) with metrics and interpretation

- GitHub repository link
- 12-minute live demo + 5-minute Q&A

### Required Deliverables:

1. **Data Preprocessing:** Feature engineering, train/test split
2. **Model Training:** At least 3 models (e.g., logistic regression, tree, Random Forest)
3. **Evaluation:** Confusion matrix, precision/recall/F1, ROC curve
4. **Interpretation:** Feature importance, explain top 3 predictions
5. **Conclusion:** Model limitations, real-world applicability

### Example Projects:

- Ghana student dropout prediction (identify at-risk students early)
- Credit scoring model for MTN Mobile Money
- Cassava disease classification (healthy vs diseased)
- Accra housing price prediction (regression)

### 4.1 Scoring Rubric (25 points)

| Criterion             | Points | Excellent (5)                                  | Good (3-4)                   | Needs Work (1-2)            |
|-----------------------|--------|------------------------------------------------|------------------------------|-----------------------------|
| <b>Data Pipeline</b>  | 5      | Clean preprocessing; no leakage                | Mostly clean; minor issues   | Data leakage or poor splits |
| <b>Model Variety</b>  | 5      | 3+ models; thoughtful selection                | 2-3 models; adequate         | 1 model or poor choices     |
| <b>Evaluation</b>     | 8      | Complete metrics; ROC/CAM plots                | Basic metrics; missing plots | Incomplete evaluation       |
| <b>Interpretation</b> | 4      | Clear feature importance; explains predictions | Basic interpretation         | Limited interpretation      |
| <b>Presentation</b>   | 3      | Professional report, clear demo                | Adequate presentation        | Disorganized or unclear     |

**Minimum Score to Pass:** 18/25

## 5 Final Demo Day: Capstone AI Solution (Week 16)

**Goal:** Students deliver a portfolio-ready AI project demonstrating all 4 phases.

**Team Size:** 2–3 students per team

### Submission Format:

- GitHub repository (code, documentation, README)
- Trained model(s) / deployed agent
- Technical report (3–5 pages)
- 10-slide presentation deck
- Demo video (optional but encouraged)
- 15-minute live presentation + 10-minute Q&A
- Team roles document (who did what)

### Required Deliverables:

1. **Problem Framing:** Real Ghana use case; why it matters
2. **Data:** Real or realistic dataset; describe source
3. **Pipeline:** End-to-end: load → clean → engineer → train → evaluate → deploy
4. **Model:** Working model or agent
5. **Evaluation:** Metrics, visualizations, test results
6. **Interpretation:** What does the model do well? Where does it fail?
7. **Deployment:** Can someone else run this code and make predictions?
8. **Documentation:** README, architecture diagram, deployment guide

### Example Projects:

- **Image Classification:** Cassava disease detection (help Ghanaian farmers diagnose crop disease)
- **NLP Agent:** Customer service chatbot for MTN or mPharma
- **Predictive Model:** Student performance forecasting (intervene early)
- **Time Series:** Electricity demand forecasting for Accra
- **Recommendation System:** Educational content recommendation for students

### 5.1 Scoring Rubric (50 points)

| Criterion              | Points | Excellent<br>(9-10)              | Good (7-8)               | Pass<br>(5-6)    | Needs Work<br>(1-4) |
|------------------------|--------|----------------------------------|--------------------------|------------------|---------------------|
| <b>Problem Framing</b> | 10     | Clear Ghana context; real impact | Good problem; minor gaps | Adequate problem | Unclear or generic  |

| Criterion                          | Points | Excellent (9-10)                                | Good (7-8)                   | Pass (5-6)                    | Needs Work (1-4)        |
|------------------------------------|--------|-------------------------------------------------|------------------------------|-------------------------------|-------------------------|
| <b>Technical Depth</b>             | 15     | Strong feature engineering; advanced techniques | Solid ML/DL; good choices    | Basic pipeline; simple models | Incomplete or shallow   |
| <b>Evaluation &amp; Metrics</b>    | 10     | Comprehensive; multiple metrics; visualizations | Good metrics; charts         | Basic metrics                 | Minimal evaluation      |
| <b>Code Quality &amp; Docs</b>     | 10     | Professional; reproducible; well-commented      | Good structure; docs present | Adequate code; some docs      | Poor structure; no docs |
| <b>Presentation &amp; Delivery</b> | 5      | Polished deck; clear demo; professional         | Good presentation; clear     | Adequate; some confusion      | Disorganized or unclear |

**Minimum Score to Pass:** 30/50 (60%)

## 5.2 Presentation Checklist

**15-minute presentation should cover:**

- ☐ Problem: What problem are you solving? Why does it matter in Ghana?
- ☐ Data: Where did you get the data? How much? What challenges?
- ☐ Model: What algorithm? Why this one? Architecture/hyperparameters?
- ☐ Results: What are your metrics? How does it perform?
- ☐ Interpretation: What did you learn? What are the limitations?
- ☐ Demo: Show the model in action on a real example
- ☐ Future: What would you improve next?

### Q&A Tips:

- “Why did you choose Random Forest over XGBoost?” → Be ready to explain trade-offs

- “How would you handle this failure case?” → Show you thought about limitations
- “Can you deploy this to production?” → Explain your architecture for serving

## 6 Assessment Logistics

### 6.1 Before Demo Day

- ☐ Students submit code and report 48 hours before
- ☐ Instructors review for basic completeness
- ☐ Flag any issues (missing files, broken code) for students to fix

### 6.2 During Demo Day

- ☐ Allocate 15 min per student (or 20 min per team)
- ☐ Use rubric to score in real-time
- ☐ Take notes on what went well and gaps to improve
- ☐ Share rubric with students after presenting

### 6.3 After Demo Day

- ☐ Average scores across all students
- ☐ Identify common struggles (feature engineering, model selection, etc.)
- ☐ Adjust next cohort’s curriculum based on findings
- ☐ Share top projects as examples for future students

## 7 Grade Distribution

For each cohort, expect roughly:

- **18%** Excellent (23–25)
- **35%** Very Good (20–22)
- **32%** Good (18–19)
- **10%** Needs Improvement (<18)

If >15% score below 18, review the phase difficulty and adjust supports.

## 8 Bonus: Extra Credit Opportunities

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Students can improve their scores by:

- **Mini Demo Days:** Deploy model to web API (Flask/FastAPI)
  - **Capstone:** Open-source contribution; submit to Kaggle competition
  - **Presentations:** Translate docs to local languages (Twi, Ga)
  - **Mentoring:** Help struggling peers with exercises
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*Rubrics designed for CodeTrain's 16-week curriculum. Adapt based on cohort level.*